

Zomato Data Analysis

Exploratory Data Analysis of Restaurant Services, Ratings, Votes and Cost

Project Type: Data Analytics / Exploratory Data Analysis

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn

Dataset: Zomato restaurant dataset (CSV)

1. Objective

The objective of this project is to explore and analyze a Zomato restaurant dataset to understand restaurant categories, online-order availability, customer ratings, customer votes and approximate cost for two people. The analysis uses exploratory data analysis and visualizations to identify patterns in restaurant services and customer engagement.

2. Dataset Overview

Attribute	Value
Rows analyzed	148
Columns	7
Numeric columns	rate, votes, approx_cost(for two people)
Categorical columns	name, online_order, book_table, listed_in(type)

The notebook output shows 148 non-null records across all seven columns, with no missing values reported by the DataFrame info output. The rate field is converted from strings such as '4.1/5' into a numeric rating.

3. Data Preprocessing

- Loaded the CSV dataset using Pandas.
- Inspected the first records to understand the structure and values.
- Converted the rate field from text format such as '4.1/5' to a floating-point value using a custom function.
- Inspected data types and non-null counts using DataFrame.info().

4. Exploratory Data Analysis

- Restaurant type distribution using a count plot.
- Total customer votes grouped by restaurant type.
- Identification of the restaurant with the maximum number of votes.
- Distribution of restaurants with and without online ordering.
- Distribution of restaurant ratings.
- Distribution of approximate cost for two people.
- Comparison of ratings for restaurants with and without online ordering using a box plot.

- Heatmap showing the relationship between restaurant type and online-order availability.

5. Key Findings

- The dataset contains 148 restaurant records and seven attributes used for the analysis.
- The notebook identifies Empire Restaurant as the restaurant with the highest number of customer votes in the analyzed dataset.
- Restaurant type and online-order availability are compared using count plots and a heatmap.
- Customer rating distribution is examined to understand how ratings are spread across the restaurants.
- Approximate cost for two people is analyzed to understand the cost distribution.
- Ratings are compared across online-order availability to explore whether rating patterns differ between the two groups.

6. Business / Analytical Interpretation

The analysis can help restaurants and food-service businesses understand service availability, customer engagement and pricing patterns. Customer votes can be treated as an indicator of engagement within this dataset, while ratings provide a measure of customer feedback. Comparing these measures across restaurant types and online-order availability can support decisions about service offerings and customer experience.

7. Recommendations

- Restaurants can monitor customer ratings and votes together rather than relying on rating alone.
- Restaurant operators can compare online-order availability with rating patterns before making service decisions.
- Cost distributions can be used to understand the price positioning represented in the dataset.
- Restaurant categories with stronger customer engagement can be examined further to understand the practices associated with that engagement.

8. Limitations and Future Scope

- The analyzed notebook contains seven fields and does not include cuisine or location columns, so cuisine- and location-level conclusions cannot be derived from the current notebook.
- The current analysis is exploratory; it does not establish causal relationships between ratings, online ordering, cost and other variables.
- Future work could add more restaurant attributes and perform statistical analysis or predictive modeling if suitable data is available.
- Additional visualizations can be added for deeper analysis of restaurant types, ratings, cost and customer engagement.

9. Conclusion

This project demonstrates a Python-based exploratory analysis workflow for Zomato restaurant data. The workflow covers data loading, preprocessing, data-quality inspection, exploratory visualization and interpretation of restaurant services, customer ratings, votes and cost. The resulting analysis provides a foundation for further business-oriented restaurant and customer analytics.